

Qingdao University of Technology
Qingdao, Shandong, China

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Editor-in-Chief
Computing

Dear Editor-in-Chief,

We submit the manuscript entitled “*Resource-Aware Permission Governance for Dependable Multi-Tenant LLM Agent Runtimes*” for consideration as a regular research article in *Computing*.

The manuscript addresses a systems dependability problem at the intersection of resource admission and permission governance. Under hostile resource pressure, a shared policy executor can miss its decision deadline; if the runtime fails open, this availability failure becomes an unsafe admission. We present Neuron, a Java runtime architecture that reserves permission capacity and couples a resource-pressure signal to the escalation policy. The central contribution is the experimentally separable distinction between capacity isolation and cross-layer coupling; identity propagation, privilege non-increase, and correlated auditing are supporting mechanisms.

The evaluation uses production Java 21 token-bucket and escalation-policy code, four predeclared architectures, independent JVM runs on Windows and an Ubuntu/KVM VPS, deadline and attacker-pressure sweeps, capacity sensitivity, process-separated permission-service fault injection, API-derived trace replay, and a post-selection holdout on six previously unused workload profiles. The frozen data and one-click replication package contain the raw JSONL observations, environment records, analysis scripts, generated values, and source code. The process-separated VPS experiment is explicitly a same-host loopback study; we do not claim WAN or multi-node behavior.

We believe the manuscript fits *Computing*’s interest in dependable, adaptive, and trustworthy computing systems because it provides a systematic runtime foundation and a controlled architectural comparison rather than a feature-only security prototype. The work is original, is not under consideration elsewhere, and has been approved by all authors. Both authors are corresponding authors. We have no competing interests and received no funding for this study.

Thank you for your consideration.

Sincerely,

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